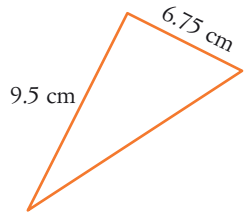
Name: _____

WORKSHEET

Using Pythagoras' theorem

1 Given that these triangles are right-angled, calculate the length of the missing sides correct to two decimal places.

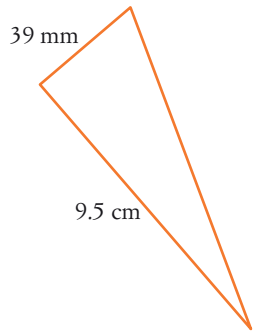
a



b

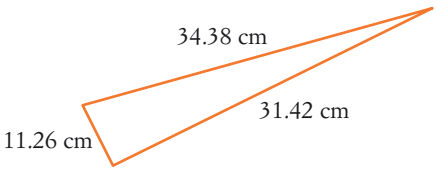


c

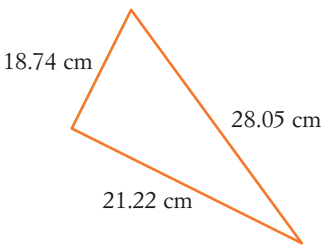


2 Use your knowledge of Pythagoras' theorem to decide if these triangles are right-angled, acute-angled or obtuse-angled.

a

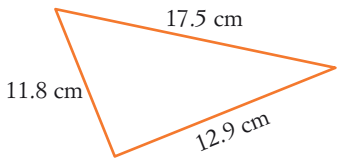


b



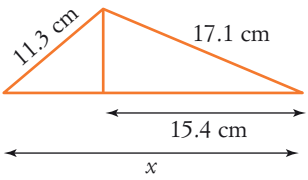
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c

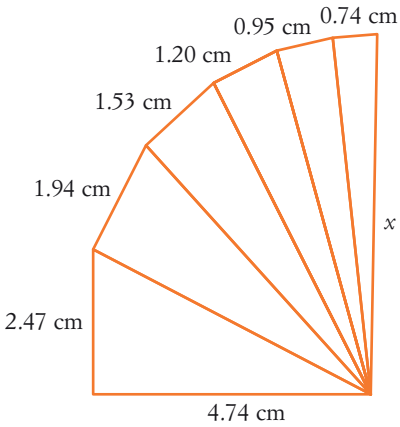


3 Find the length of the side marked x in each of these figures correct to two decimal places.

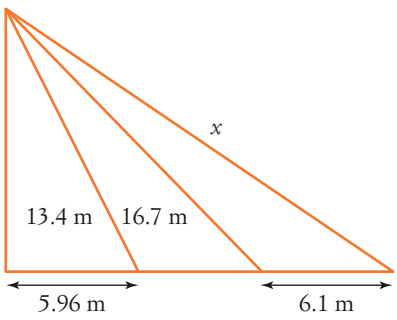
a



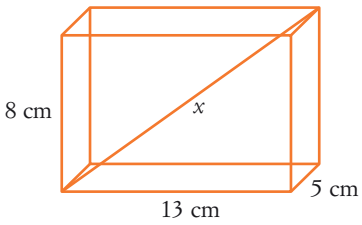
b



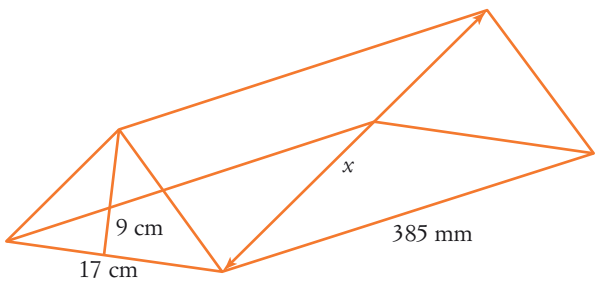
c



d

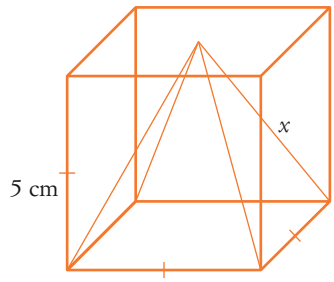


e



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f



4 Use Pythagoras' theorem to solve each of these problems, correct to two decimal places.

a A 3.6 m ladder needs to reach the top of a wall that is 2.9 m high. What is the minimum distance the ladder needs to be placed from the bottom of the wall?

b A 6.4 m flagpole needs a guide wire to attach at 80% of its height. If the guide wire will be tethered 2.3 m from the base, how long must it be?

c An ice-cream cone has a diameter of 7 cm and a slant length of 16 cm. If the cone has chocolate on the inside of the bottom 3 cm (of the height) of the cone, what percentage of the slant length of the cone is covered internally with chocolate?

d A 1.8 m man is standing on boat that sits 2.3 m above sea level. He is looking at his friend (who is 1.75 m tall) on a cliff. They have between them a thin piece of rope that measures 72 m, and the boat is sitting 45 m from the base of the cliff. How tall is the cliff?

e A 6.9 m tree has a 13.2 m guide wire attached to its trunk 2.3 m below the top of its canopy. A second guide wire is to be added, this time attached 4.1 m from the ground. If the second guide wire is 10.6 m long, what is the distance between the two guide wires?

f A 10 cm square piece of paper is folded diagonally five times. What are the dimensions of the final resulting triangle?

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Answers

1 **a** 11.65 cm

b 4.73 cm

c 10.27 cm

2 **a** obtuse-angled

b acute-angled

c right-angled

3 **a** 23.91 cm

b 6.13 cm

c 21.40 cm

d 16.06 cm

e 40.49 cm

f 6.12 cm

4 **a** 2.13 m

b 5.61 m

c 19.74%

d 58.55 m

e 2.60 m

f $\frac{5}{2}$ cm, $\frac{5}{2}$ cm, $\frac{5\sqrt{2}}{2}$ cm

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